

		Since v is a constant, ω is proportional to $\frac{1}{r}$.
11	C	$\frac{GMm}{R^2} = mR\omega^2 = mR\left(\frac{2\pi}{T}\right)^2$ $T^2 = \left(\frac{4\pi^2}{GM}\right)R^3, \text{ thus } \left(\frac{R_x}{R_y}\right)^3 = \left(\frac{T_x}{T_y}\right)^2 = \left(\frac{4}{1}\right)^3$ $\frac{R_x}{R_y} = \left(\frac{4}{1}\right)^{\frac{3}{2}} = 8$
12	B	Using $P = \frac{1}{2} \rho v_{rms}^2$